# Stochastic Optimization for E-commerce Inventory Management Under Uncertain Customer Demand

**Course:** 21AIE321T · Stochastic Decision Making
**Assignment:** IV (Mini Project)
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## Abstract

An online seller must decide how many units of a product to buy before customer demand is known. This project models that decision for a single product (wireless headphones) as a one-period stochastic optimization problem. The decision variable is the integer order quantity Q between 0 and 500, and the objective is to maximize expected profit under random demand. Expected profit is estimated by Monte Carlo simulation with 10,000 demand scenarios, and the best Q is found by exhaustive search over all 501 candidate values. The result is compared with a deterministic model that assumes demand is exactly 120 units, using independent test scenarios that were not used to choose Q. In the base case (Poisson demand with mean 120) both models order 120 units, so the two policies tie. The deterministic model, however, predicts a profit of ₹84,000, whereas the average profit on test scenarios is ₹76,151. Sensitivity experiments on mean demand, demand variability and cost parameters show that higher variability lowers profit and sharply increases downside risk while moving the best order quantity only slightly (from 120 to 117), and that cost changes shift the best quantity by a few units, with stochastic gains of at most 0.65% of profit. All costs and demand data are illustrative assumptions and synthetic; they are not real company data.

## 1. Introduction and Problem Statement

Online retailers must commit to a purchase quantity before they observe customer demand. Ordering too much leaves unsold stock that carries holding costs and ties up money. Ordering too little leaves demand unmet, which incurs a shortage penalty and can harm customer service. This project studies that trade-off for one product, wireless headphones, sold over a single sales period.

**Problem statement.** Choose the order quantity Q, an integer from 0 to 500, that maximizes the expected profit when the demand D in the sales period is a random variable.

**Objectives.**

1. Formulate a stochastic profit model with explicit revenue, purchasing, holding and shortage terms.
2. Estimate expected profit by Monte Carlo simulation and find the best Q by exhaustive search.
3. Compare the stochastic solution with a deterministic model that assumes demand equals its mean, using independent test scenarios.
4. Analyse sensitivity to the mean demand, the demand variability and the cost parameters.
5. Relate the findings to SDG 9 (Industry, Innovation and Infrastructure).

**Scope.** One product, one period, no carry-over stock, no lead times and no supplier limits. All parameter values are illustrative assumptions and all demand data are synthetic.

## 2. Background

The problem is a form of the newsvendor problem: a single-period, buy-once decision in which the best quantity depends on the balance between the cost of ordering one unit too many (overage cost, Co) and one unit too few (underage cost, Cu). The optimal quantity is the smallest Q for which $P(D \leqslant Q) \geqslant Cu / (Cu + Co)$ (Edgeworth, 1888; Arrow, Harris and Marschak, 1951; Silver, Pyke and Thomas, 2017). In this project $Cu = p - c + s =$ ₹900 and $Co = c + h =$ ₹900, so the critical ratio is 0.5 and the best Q is the median of demand.

When the expected value of a profit function cannot be handled analytically, or when one wants a method that extends to other models, it can be approximated by averaging the profit over simulated scenarios. This is Monte Carlo simulation, also called sample average approximation in stochastic programming (Shapiro, Dentcheva and Ruszczyński, 2009; Kleywegt, Shapiro and Homem-de-Mello, 2002). The standard error of such an average shrinks in proportion to $1/\sqrt{N}$, where N is the number of scenarios (Law, 2015).

A common shortcut is to plan with a single average demand figure. This ignores variability, so it can misjudge both the expected profit and the risk. Section 7 quantifies this for the present model.

## 3. Assumptions and Parameters

Table 1 lists the model parameters. They are hypothetical values chosen for illustration; they are not real company statistics.

**Table 1. Model parameters (illustrative assumptions).**

| Parameter | Symbol | Value | Meaning |
|---|---|---|---|
| Selling price per unit | p | ₹1,500 | Revenue for each unit sold |
| Purchasing cost per unit | c | ₹800 | Cost of buying one unit before demand is known |
| Holding cost per unsold unit | h | ₹100 | Cost of storing and tying up money in each unsold unit |
| Shortage penalty per unmet unit | s | ₹200 | Extra service or reputation penalty for each unit of demand not served |
| Maximum inventory capacity | Qmax | 500 units | Upper limit on the order quantity |
| Mean customer demand | μ | 120 units | Average demand per sales period |

The shortage penalty s is an additional per-unit penalty. It is not lost sales revenue, because the revenue term already counts only the units actually sold, so no revenue is double-counted. Unsold units are assumed to have no salvage value.

## 4. Mathematical Model

**Decision variable.** Q, the number of units purchased before demand is known. Q is a non-negative integer.

**Uncertain variable.** D, the actual customer demand in the sales period.

**Profit function.**

$$\text{Profit}(Q, D) = p \cdot \min(Q, D) - c \cdot Q - h \cdot \max(Q - D, 0) - s \cdot \max(D - Q, 0)$$

The four terms are the revenue from units sold, the purchasing expenditure, the holding cost on unsold units, and the shortage penalty on unmet units.

**Worked example.** With Q = 120 and D = 100, the profit is 1500·100 − 800·120 − 100·20 = 150,000 − 96,000 − 2,000 = ₹52,000. With D = 130 it is 1500·120 − 800·120 − 200·10 = ₹82,000. When D = 120 exactly, there is no unsold stock and no shortage, and the profit is ₹84,000, the maximum possible for Q = 120.

**Stochastic optimization problem.**

$$\text{maximize } E[\text{Profit}(Q, D)] \text{ subject to } 0 \leqslant Q \leqslant 500, \ Q \text{ integer.}$$

The quantity E[Profit(Q, D)] is the expected profit: the long-run average over all possible demand outcomes. It differs from the profit in one scenario, which is what happens in a single sales period and can be much higher or lower.

**Monte Carlo approximation.** With N simulated demand scenarios $D_1, \ldots, D_N$,

$$E[\text{Profit}(Q, D)] \approx (1/N) \Sigma \text{Profit}(Q, D_i),$$

with standard error $s_P/\sqrt{N}$, where $s_P$ is the sample standard deviation of the N profits. We use N = 10,000.

## 5. Demand Model and Solution Method

**Demand distribution.** The base case uses D ~ Poisson(λ = 120). A Poisson distribution models non-negative integer counts, and its mean and variance are both equal to λ, so the base-case standard deviation is about 11 units. This is an assumption, not a claim that Poisson is the best real-world demand model. Its main limitation is that variance is tied to the mean, which rules out more variable demand.

For the variability experiments the mean is kept fixed at 120 and demand follows a negative binomial distribution with variance 240, 480 and 960. Given a target mean μ and variance $\sigma^2$ (with $\sigma^2 > \mu$), the NumPy parameters are $p = \mu/\sigma^2$ and $n = \mu^2/(\sigma^2 - \mu)$; for example $\sigma^2 = 480$ gives n = 40 and p = 0.25. The theoretical mean and variance were verified for each case so that only variability changes.

**Synthetic data.** Demand scenarios were generated with fixed random seeds: seed 42 for the training scenarios that choose Q, and seed 2026 for independent test scenarios used only to compare policies. The generated Poisson sample (10,000 values) has sample mean 120.02 and sample variance 118.3; the negative binomial samples have variances 242.4 and 464.8 against targets of 240 and 480, differences that are due to sampling. Figure 1 shows the simulated demand.

[Insert Figure 1 here: report/figures/01_demand_histogram.png]

**Figure 1. Histogram of 10,000 simulated Poisson(120) demand scenarios (synthetic).**

**Why exhaustive search.** Q is a single integer with only 501 possible values, so the expected profit of every Q can be computed and compared directly. This guarantees the best Q on the simulated scenarios, with no risk of a local optimum and no need for a solver.

**Deterministic baseline.** The deterministic model assumes demand is exactly 120 in every period, so Profit(Q, 120) is a function of Q alone and its maximizer is found by the same search.

**Fair comparison.** Q is chosen on the training scenarios. Both policies are then evaluated on the same independent test scenarios. Because both policies face the same scenarios, the profit difference is computed scenario by scenario (a paired comparison) and reported with an approximate 95% interval (1.96 times the standard error of the differences).

## 6. Implementation and Validation

The model is implemented in Python (NumPy, pandas, SciPy, Matplotlib) in a virtual environment. Reusable code is in a `src` package (profit function, demand generation, simulation, optimization, deterministic model, evaluation and sensitivity experiments), and the analysis is presented in Jupyter notebooks, with the notebooks importing the functions from `src`.

Validation was done in several ways:

- The profit function was checked against hand calculations, including a case that confirms the shortage penalty is not double counted.
- Demand values were checked to be non-negative integers, reproducible for a fixed seed, and the negative binomial parameters were checked to give the intended mean and variance.
- The vectorized optimizer was compared with a plain loop, and the Poisson optimum was confirmed by an exact calculation using the Poisson probabilities.
- The convergence of the Monte Carlo estimate was examined: at Q = 120 the standard error fell from ₹911.7 to ₹304.5 to ₹94.8 as N grew from 100 to 1,000 to 10,000, roughly a factor of $\sqrt{10}$ each time.
- A suite of 19 unit tests passes. One test initially failed because of an arithmetic mistake in its expected value (₹30,000 instead of ₹35,000), not a defect in the model, and the expected value was corrected.

## 7. Results: Base Case and Deterministic Comparison

**Stochastic optimum.** Evaluating every integer Q from 0 to 500 on the training scenarios gives Q* = 120, with estimated expected profit ₹76,196 (standard error ₹94.8). The newsvendor critical ratio is 900/(900 + 900) = 0.5, so theory also gives the Poisson(120) median, 120. An exact calculation with the Poisson probabilities also gives Q = 120 and an expected profit of ₹76,139, which is within one standard error of the simulated value. Figure 2 shows expected profit against Q. The curve peaks at 120 and falls by about ₹900 per unit beyond it, because each additional unsold unit costs c + h = ₹900.

[Insert Figure 2 here: report/figures/03_inventory_vs_expected_profit.png]

**Figure 2. Expected profit against order quantity (left: Q = 0 to 500; right: zoom near the optimum with an approximate 95% interval).**

**Deterministic optimum.** With demand fixed at 120, the best order is also 120 and the predicted profit is ₹84,000. The deterministic model therefore recommends the same quantity as the stochastic model in the base case.

**Independent test comparison.** Table 2 evaluates the common order quantity on 10,000 independent test scenarios. The two policies are identical, so the profit difference is exactly zero. The key difference is in what each model predicts: the deterministic model predicts ₹84,000, but the average test profit is ₹76,151, so it overstates profit by about ₹7,849 (about 10%). It also says nothing about risk: a stockout occurs in 47.5% of periods, and in the worst 5% of periods the profit is ₹55,200 or less.

**Table 2. Base-case results on 10,000 independent test scenarios (Q = 120, Poisson demand).**

| Metric | Value |
|---|---|
| Average profit | ₹76,151 (standard error ₹95) |
| Median profit | ₹80,800 |
| 5th percentile profit | ₹55,200 |
| 95th percentile profit | ₹83,800 |
| Average unsold units | 4.35 |
| Average unmet demand | 4.42 units |
| Stockout frequency | 0.475 |
| Deterministic predicted profit | ₹84,000 |

[Insert Figure 3 here: report/figures/05_policy_profit_distribution.png]

**Figure 3. Profit across 10,000 independent test scenarios for both policies (the two histograms coincide because both order 120 units).**

The profit distribution is skewed to the left with an upper limit of ₹84,000, the profit when demand equals exactly 120. This asymmetry shows that the average profit is not what happens in a typical period, and that downside risk is larger than the average suggests.

# 8. Sensitivity Analysis

Each experiment changes one input and keeps the others at their base values. In every case Q* was chosen on training scenarios and compared with the deterministic order on independent test scenarios.

## 8.1 Experiment A: mean demand

For mean demand 80, 100, 120, 150 and 180, the best order quantity equals the mean demand in every case, and the expected test profit rises from ₹49,596 to ₹116,375, about ₹668 per additional unit of mean demand (less than the ₹700 unit margin because of demand uncertainty). The deterministic order equals the mean by construction, so the policies tie. The largest simulated demand (229) stayed far below Qmax = 500, so the capacity constraint was never binding.

[Insert Figure 4 here: report/figures/07_effect_of_mean_demand.png]

**Figure 4. Effect of mean demand on the optimal order quantity and the expected profit.**

## 8.2 Experiment B: demand variability (mean fixed at 120)

Table 3 shows that raising the variance from 120 to 960 while keeping the mean at 120 lowers the average profit from ₹76,151 to ₹61,466 and the 5th percentile profit from ₹55,200 to ₹11,500, yet moves the best order quantity only from 120 to 117. The deterministic model predicts ₹84,000 in all cases, so its prediction error grows from ₹7,849 to ₹22,534. The stochastic order is better than the deterministic order by ₹26.8 (± 17.6) at variance 480 and ₹102.4 (± 52.3) at variance 960, which are small (below 0.2% of profit) though distinguishable from zero. Variability matters even though the mean is unchanged because profit is a non-linear function of demand: shortages and overstock both reduce profit, so a wider spread of demand means more periods on the costly sides of Q. At Q = 120 the average number of unsold units rises from 4.35 (3.6% of the order) at variance 120 to 12.63 (10.5%) at variance 960.

**Table 3. Effect of demand variability (mean 120; test scenarios).**

| Demand | Variance | Q* | Q (deterministic) | Average profit (₹) | 5th percentile profit (₹) | Gain over deterministic (₹, ± 95%) |
|---|---|---|---|---|---|---|
| Poisson | 120 | 120 | 120 | 76,151 | 55,200 | 0.0 ± 0.0 |
| Negative binomial | 240 | 120 | 120 | 72,738 | 44,000 | 0.0 ± 0.0 |
| Negative binomial | 480 | 119 | 120 | 68,164 | 30,500 | 26.8 ± 17.6 |

| Demand | Variance | Q* | Q (deterministic) | Average profit (₹) | 5th percentile profit (₹) | Gain over deterministic (₹, ± 95%) |
|---|---|---|---|---|---|---|
| Negative binomial | 960 | 117 | 120 | 61,466 | 11,500 | 102.4 ± 52.3 |

[Insert Figure 5 here: report/figures/08_effect_of_demand_variability.png]

**Figure 5. Effect of demand variance (mean 120) on the optimal order quantity (left) and on profit (right).**

## 8.3 Experiment C: cost parameters

The purchasing cost c, holding cost h and shortage penalty s were varied one at a time (Table 4). The best quantity moves in the directions predicted by the critical ratio: it falls as c rises (123 to 117) and as h rises (121 to 118), and it rises as s rises (118 to 124). The simulated Q* equals the newsvendor quantity in 14 of the 15 cases and differs by one unit in the other (s = 800), which is expected because the profit curve is flat near its peak. Each ₹100 increase in c reduces expected profit by roughly ₹12,100 (about 120 units × ₹100). The deterministic order stays at 120 in every case, so wherever Q* differs from 120 the stochastic policy earns slightly more: the gain lies between ₹7 and ₹460 (at most 0.65% of profit). Only the case h = 0 is not statistically distinguishable from zero. The changes are small because Poisson(120) demand has a standard deviation of only about 11 units and the profit curve is flat near its maximum.

**Table 4. Effect of cost parameters (Poisson(120) demand; test scenarios).**

| Parameter | Value (₹) | Critical ratio | Q* | Q (deterministic) | Gain (₹, ± 95%) |
|---|---|---|---|---|---|
| c | 600 | 0.611 | 123 | 120 | 274.0 ± 50.9 |
| c | 700 | 0.556 | 121 | 120 | 54.5 ± 17.6 |
| c | 800 (base) | 0.500 | 120 | 120 | 0.0 |
| c | 900 | 0.444 | 118 | 120 | 91.3 ± 34.6 |
| c | 1000 | 0.389 | 117 | 120 | 339.2 ± 50.9 |
| h | 0 | 0.529 | 121 | 120 | 7.0 ± 16.6 |
| h | 50 | 0.514 | 120 | 120 | 0.0 |
| h | 100 (base) | 0.500 | 120 | 120 | 0.0 |
| h | 200 | 0.474 | 119 | 120 | 26.1 ± 18.6 |
| h | 400 | 0.429 | 118 | 120 | 173.2 ± 40.4 |
| s | 0 | 0.438 | 118 | 120 | 103.4 ± 30.7 |
| s | 100 | 0.471 | 119 | 120 | 28.6 ± 16.7 |

| Parameter | Value (₹) | Critical ratio | Q* | Q (deterministic) | Gain (₹, ± 95%) |
|---|---|---|---|---|---|
| s | 200 (base) | 0.500 | 120 | 120 | 0.0 |
| s | 400 | 0.550 | 121 | 120 | 49.4 ± 19.6 |
| s | 800 | 0.625 | 124 | 120 | 459.6 ± 88.8 |

[Insert Figure 6 here: report/figures/09_effect_of_cost_parameters.png]

**Figure 6. Effect of the purchasing cost, holding cost and shortage penalty on the optimal order quantity (top) and the expected profit (bottom).**

# 9. Discussion: SDG Relevance, Limitations and Extensions

## 9.1 SDG 9: Industry, Innovation and Infrastructure

The model supports SDG 9 in four ways, each tied to a result. First, it supports more efficient business operations by choosing an order quantity that maximizes expected profit (Q* = 120, expected profit about ₹76,151). Second, it supports data-driven decisions: the order is based on simulated scenarios rather than a single guess, and doing so shows that the deterministic plan overstates profit by about ₹7,849. Third, it supports better resource use, since ordering close to the demand level limits stock that is bought but not sold (about 4.35 units, or 3.6% of the order, in the base case). Fourth, it supports resilient supply-chain processes by measuring downside risk: the 5th percentile profit falls from ₹55,200 to ₹11,500 as variability rises, information a single-number forecast cannot provide. The method (model demand, simulate, optimize, test on new scenarios) is also reusable for other products, which reflects the innovation theme of the goal. SDG 9 contains formal targets; the exact wording should be taken from the UN SDG website (United Nations, n.d.-a).

## 9.2 SDG 12: Responsible Consumption and Production (with caution)

Unsold stock is a form of surplus production. At Q = 120, the average number of unsold units per period rises from 4.35 to 12.63 as demand variance rises from 120 to 960. This suggests that reducing demand variability, for example through better forecasting, matters more for surplus than changing the optimizer, because the best order quantity moves by only three units. The model counts units and does not measure environmental impact, so no environmental benefit is quantified (United Nations, n.d.-b).

### 9.3 Practical implications

A small online seller can apply the same workflow to set an order quantity for each product and period, and to any buy-once, sell-later product. If costs are symmetric and demand is fairly predictable, ordering near the average demand is nearly optimal. Uncertainty affects profit and risk much more than it affects the order size, so a business should budget for lower profit and a larger downside than the deterministic plan implies. When shortages are costly or purchasing is cheap, ordering slightly above the mean pays off, and the opposite holds when holding or purchasing costs are high.

### 9.4 Limitations

- All costs and demand data are synthetic assumptions, not real company data.
- The model covers one product and one period, with no carry-over stock, lead times or supplier limits, and no salvage value for unsold units.
- The demand distributions (Poisson and negative binomial) are assumed and were not fitted to real sales history.
- Q* was chosen from a single training sample, so another seed could give a neighbouring value; the profit curve is flat near its peak.
- The shortage penalty is a single number that is hard to measure in practice.
- Differences between the policies are small and should not be overstated.

### 9.5 Possible extensions

Fitting the demand distribution to real sales data, multi-period or seasonal ordering with carry-over stock, several products sharing a budget, price as a decision variable, a risk-averse objective such as maximizing a low percentile of profit, and uncertainty in supplier lead times.

## 10. Conclusion

This project formulated an e-commerce order-quantity decision as a stochastic optimization problem and solved it with Monte Carlo simulation and exhaustive search. In the base case the stochastic and deterministic models recommend the same order of 120 units, so stochastic optimization did not improve the decision itself. Its value was in the evaluation: the deterministic model overstated profit by about ₹7,849 and could not describe risk, while the stochastic model reported realistic profit, stockout frequency and downside. Variability lowered profit and raised risk far more than it changed the best order quantity, and cost asymmetry shifted the best quantity by a few units with small gains of at most 0.65% of profit. These conclusions rest on illustrative synthetic data and should be re-checked with real sales history before any real use.

# References

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